

Roll Angle Stabilization of Fixed-wing UAVs in occurrence of noises by using PID with EKF controller

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Abstract:

In recent years there is tremendous growth of research focused on unmanned aerial vehicle stabilization and its control. This research article focused on control system design for the stabilization of Roll angle of Fix –wing UAV. There are several different types of UAVs available in the market, but fixed-wing UAVs is the most common one which is used for surveillance and rescue applications. Like other UAVs, Fix-wing UAVs are also non linear in nature and its stabilization during flight is most critical task. In lateral autopilot controlling of Fixing UAV the roll angle stabilization play very significant part and its proper controlling is essential for UAV stability. In this research work PID with Extended kalman Filter control algorithm is proposed for stabilization of roll angle of fixed-wing UAV under uncertainty conditions. PID parameters are tuned in such a way that the roll stabilized as fast as possible and within limited overshoot and the addition of extended kalman filter (EKF) makes UAV more stable by canceling the noises produced by the system and sensors. Simulation results proved that, the proposed control technique works well and produce effective results for fixed-wing UAV's roll angle stabilization.

Keywords: UAV, Longitude Control, Latitude Control, PID, Extended Kalman Filter.

1. Introduction

Unmanned Aerial Vehicle (UAV) have recently been used with great success for military intelligence by providing a viable alternative to manned aircraft due to their smaller size, reduced risk to life, and reduced cost. Forces use UAVs in applications such as border patrolling, security Intelligence, surveillance, target acquisition and cost guards. Besides military applications of UAVs it also use in many civil applications such as search &

rescue missions, explorations, security & surveying of oil pipe lines, Forests on fire, agricultural applications and power & nuclear plants inspection.

There exists many types of UAVs, this article focuses only on fixed wing propelled UAV. The UAV is a six DOF system and its dynamics can be categorized into longitudinal and lateral dynamics. The fixed-wing UAV is nonlinear and complex in nature. It requires a suitable controller for the stability during flight. The main idea of the paper is to present a control technique for roll stabilization of UAV.

2. Related Work:

Since the advent of engineering, simulation of physical systems has been regarded as great means to predict and understand system's behavior. The engineering flight simulators reduce lifecycle costs of design because development and testing of complex aircraft systems can be done before actual flight tests. A simulation model built to analyze performance, stability characteristics and mission of a flight vehicle. In recent years, UAVs have been researched immensely due their complex aerodynamics and constraints in kinematics. However, some researchers came up with effective solutions in this area. In 2008, Stephen Jackson used sliding mode technique for control of small sized fixed-wing UAV (S. Jackson 2008). In 2012, Yang Chen applied fuzz logic controller (Yang Chen 2012). In 2009, soha pouya applied pole-placement technique for longitudinal control of UAV (Pouya, S 2009). This article represents the mathematical modeling of a fixed-wing UAV with an auto tuned PID controller for roll angle stabilization.

The organization of the paper is structured as the section III refers to the mathematical modeling, which follows the discussion on Kinematics and Dynamics of fixed-wing UAV. Section IV is regarding the proposed Control technique. Experimental results and simulations have been provided in section V.

3. Mathematical Modeling of fixed-wing UAV:

A fixed-wing aircraft is capable of flight using fixed wings that generate lift caused by the vehicle's forward thrust/velocity and the shape of the wings. For flight dynamics it is necessary to specify an inertial frame of reference because Newton laws of motion are only valid when acceleration is measured with respect to inertial frame which is actually at rest in the universe. After all other system is useful in flight dynamics is an axes system fixed at the centre of earth which is called body frame as shown in figure 1.

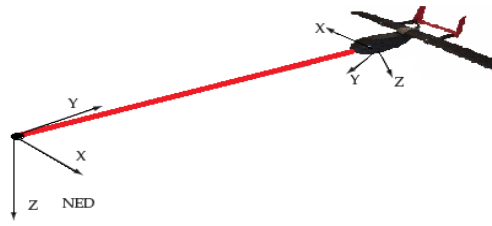


Figure 1: Body and NED frame of Fixed-wing UAV

To be able to relate the rotation and motion in the Body frame to the (North, East and down) NED frame a rotation matrix is used:

$$R_B^n = \begin{bmatrix} c_\psi c_\theta & -s_\psi c_\theta + c_\psi s_\theta s_\phi & s_\psi s_\theta + c_\psi s_\theta c_\phi \\ s_\psi c_\theta & c_\psi c_\theta + s_\psi s_\theta s_\phi & -c_\psi s_\theta + s_\psi s_\theta c_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \quad (1)$$

The velocity vector v_0^b in the Body frame is decomposed in the NED-frame as:

Six degree of freedom (6 DOF) Rigid- Body forces and moments in the BODY frame are defined as:

$$\bar{f}_{RB} = \begin{bmatrix} f_0^b \\ m_0^b \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \\ L \\ M \\ N \end{bmatrix} = \begin{bmatrix} \text{forward force} \\ \text{Sideways force} \\ \text{Vertical force} \\ \text{Roll moment} \\ \text{Pitch moment} \\ \text{yaw moment} \end{bmatrix} \quad (2)$$

Where f_0^b and m_0^b are force and moments which are the sum of aerodynamic (with disturbance), gravitational and actuator forces and moments.

Equation (3) is the expression, which helps to define Six degree of freedom (6 DOF) for fixed wing aerial craft:

$$M_{RB} \dot{v} + C_{RB}(v)v + g(\eta) = \bar{f} + \omega \quad (3)$$

Where,

M = System Inertial Matrix

$C(v)$ = Coriolis Centripetal matrix

$D(v)$ = Damping Matrix

$g(\eta)$ = Vector of Gravitational forces and moments

$\bar{i}$ = Vector of Control Inputs

g_o = Vector use for pre trimming

ω = Vector of Disturbances

Expression of equation (3) is evaluated as shown in equation (4), Component form can be expressed as:

$$\left. \begin{aligned} X &= m (\dot{U} + QW - RV + g \sin \theta) \\ Y &= m (\dot{V} + UR - PW - g \cos \theta \sin \phi) \\ Z &= m (\dot{W} + PV - QU - g \cos \theta \cos \phi) \\ L &= I_x \dot{P} - I_{xz} (\dot{R} + PQ) + (I_z - I_y)QR \\ M &= I_y \dot{Q} - I_{xz} (P^2 - R^2) + (I_x - I_z)PR \\ N &= I_z \dot{R} - I_{xz} \dot{P} + (I_y - I_x)PQ + I_{xz} QR \end{aligned} \right\} \quad (4)$$

Equation (4) represents forces and moment's equation governs the motion of an aircraft with respect to an inertial frame.

3.1. Roll Analysis

A roll motion is an up and down movement of the wings of the aircraft, in roll motion aircraft rotates about its center of gravity, a point which is the average location of the mass of the aircraft. The rolling motion is being caused by the deflection of the aileron of the aircraft. For a fixed-wing UAV flying at a constant altitude, a roll angle command is often used for heading control. So for controlling heading rate it is necessary to control roll angle. The lateral equation of motion for aileron control is given by equations below (Bandu N).

$$\begin{aligned} (m_1 s - b C_{y\beta} s - C_{y\beta}) \Delta \bar{\beta}(s) - (b C_{yp} s + C_{y\phi}) \Delta \bar{\Phi}(s) + (m_1 - b C_{yr}) s \Delta \bar{\Psi}(s) &= C_{y\delta_a} \Delta \bar{\delta_a}(s) \\ (-C_{l\beta} - b C_{l\beta} s) \Delta \bar{\beta}(s) + (-C_{lp} b s - I_{xl} s^2) \Delta \bar{\Phi}(s) - (C_{lr} b s + I_{xz} s^2) \Delta \bar{\Psi}(s) &= C_{l\delta_a} \Delta \bar{\delta_a}(s) \\ (-C_{n\beta} - b C_{n\beta} s) \Delta \bar{\beta}(s) + (-C_{np} b s - I_{xz} s^2) \Delta \bar{\Phi}(s) + (-C_{nr} b s + I_{zl} s^2) \Delta \bar{\Psi}(s) &= C_{n\delta_a} \Delta \bar{\delta_a}(s) \end{aligned}$$

Where,

C_x, C_z, C_m are aerodynamic coefficient.

β, Φ, Ψ longitudinal variable.

s is reference area.

c is mean aerodynamic chord.

b is wing span.

C_l , C_n used to define the pitch and yaw moment coefficients.

C_m is the mean aerodynamic chord coefficient.

$C_{y\{\phi,\beta,p,r\}}$, $C_{l\{r,\beta,p\}}$, $C_{n\{r,\beta,p\}}$ are stability derivatives.

$C_{m\delta_e}$, C_{mq} are control derivatives.

Assuming that an aileron is the only lateral control can be expressed as in equation (5).

After solving above equations, transfer function of roll system.

$$\frac{\Delta \bar{\Phi}(s)}{\Delta \delta_a(s)} = \frac{\begin{vmatrix} (m_1 s - b C_{y\beta} s - C_{y\beta}) & C_{y\delta_a} & (m_1 - b C_{yr}) s \\ (-C_{l\beta} - b C_{l\beta} s) & C_{l\delta_a} & -(C_{lr} b s + I_{xz} s^2) \\ (-C_{n\beta} - b C_{n\beta} s) & C_{n\delta_a} & -(C_{nr} b s - I_{zl} s^2) \end{vmatrix}}{\Delta_{lat}(s)} \quad (5)$$

$$\Delta_{lat}(s) = \begin{vmatrix} (m_1 s - b C_{y\beta} s - C_{y\beta}) & -(b C_{yp} s + C_{y\phi}) & s(m_1 - b C_{yr}) \\ (-C_{l\beta} - b C_{l\beta} s) & -(C_{lp} b s + I_{xl} s^2) & -(C_{lr} b s + I_{xz} s^2) \\ (-C_{n\beta} - b C_{n\beta} s) & -(C_{np} b s + I_{xz} s^2) & -(C_{nr} b s - I_{zl} s^2) \end{vmatrix}$$

After solving and converting equation (5) into Laplace, We get,

$$\frac{\Delta \bar{\Phi}(s)}{\Delta \delta_a(s)} = \frac{k_1 s^3 - k_2 s^2 + k_3 s}{k_4 s^5 + k_5 s^4 + k_6 s^3 + k_7 s^2 + k_{10} s} \quad (6)$$

Where $k_-(\dots)$ are coefficients of UAV.

4. Proposed Control Algorithm

The main purpose of this research is to design controller which maintains the stability of a Fixwing unmanned aircraft's Roll movement according to the desired input. PID controller gives good results in controllability of roll movement, but it has certain limits as the control algorithm is not good enough to work under noisy conditions that are caused by the sensors or system, so for the elimination of noises Extended Kalman Filter (EKF) is proposed with PID controller in this research work. Figure 2 shows the overall block

diagram of Proposed Control Algorithm for controlling the movement of Roll Angle. In this diagram EKF is used for the eliminating feedback sensor noises and auto-tuned PID controller is used for getting desired response form Roll Angle of Fix-wing UAV system model.

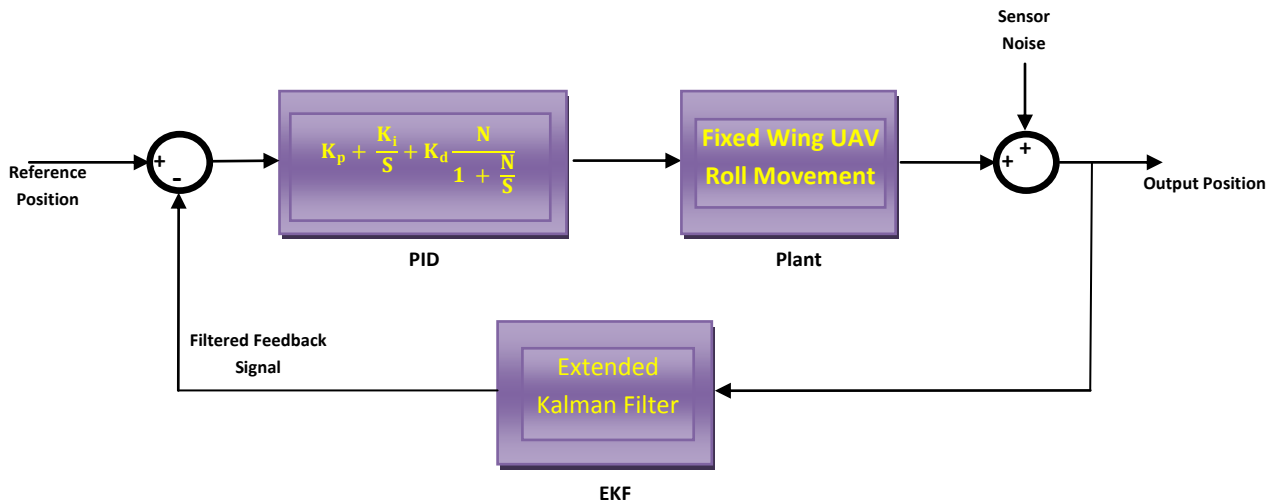


Figure 2: Block Diagram of Feedback Control System with EKF

To get the desired roll angle movement, the error signals “e” for PID control is:

$$e_{\bar{\phi}} = \bar{\phi}_{\text{ref}} - \bar{\phi}$$

And the PID controller equation can be written as:

$$U(s) = K_p e(s) + \frac{K_i}{s} e(s) + K_d \frac{N}{1 + \frac{N}{s}} e(s) \quad (7)$$

The parameters of PID controller are chosen by Auto-tuning method with optimum setting.

Applying PID controller to the system model of roll-angle, Product of equation (6) and equation (7) gives open loop system shows in equation (8)

$$\frac{\text{Output}}{\text{Input}} = \frac{k_1 s^3 - k_2 s^2 + k_3 s}{k_4 s^5 + k_5 s^4 + k_6 s^3 + k_7 s^2 + k_8 s} \cdot K_p e(s) + \frac{K_i}{s} e(s) + K_d \frac{N}{1 + \frac{N}{s}} e(s) \quad (8)$$

Finally, the open loop system equation of plant can easily be converted to the close loop feedback system by $(\frac{G}{1+GH})$.

4.1. Extended Kalman Filter:

Fixed-wing UAV systems always gives noisy output signal due to noisy signal feedback, so for the estimation of the original output and cancel the effect of noise EKF (Extended Kalman Filter) technique is used (M.Hassan Tanveer 2013).

Equation (9) is standard state space model of a nonlinear system,

$$\left. \begin{aligned} x_{k+1} &= f(x_k, u_k) + w_k \\ \bar{y}_k &= h(x_k) + v_k \end{aligned} \right\} \quad (9)$$

‘ x_k ’ is a state vector, ‘ $\bar{y}_k$ ’ is a measured process output, ‘ w_k ’ and ‘ v_k ’ are the process and measurement noises respectively. $f(\cdot)$ and $h(\cdot)$ are generic nonlinear functions. The standard kalman filter, the EKF also uses two step prediction and correction algorithm. The update equations of EKF are shown in equation (10).

$$\left. \begin{aligned} \hat{x}_{k+1} &= f(\hat{x}_k, u_k) \\ P_{k+1} &= A_k P_k A_k^T + Q \end{aligned} \right\} \quad (10)$$

Where ‘ $\hat{x}_{k+1}$ ’ is update state estimate. The time update equations project the state and covariance estimate ‘ P_{k+1} ’ from previous time step ‘ k ’ to the current time step ‘ $k+1$ ’ (M. Kamran Joyo 2013). The measurement update equations of EKF are:

$$\begin{aligned} K_k &= P_k C_k^T (C_k P_k C_k^T + R)^{-1} \\ \hat{x}_{k+1} &= f(\hat{x}_k, u_k) + K_k [\bar{y}_k - h(\hat{x}_k)] \\ P_{k+1} &= A_k (1 - K_k C_k) P_k \end{aligned}$$

Where ‘ K ’ is the correction Kalman gain vector and ‘ A_k ’ and ‘ C_k ’ cannot be used directly. With this type of limitation either Taylor series is applied or Jacobian is used. The Jacobians are defined as.

$$f'(x) = \begin{bmatrix} \frac{df_1}{dx_1} & \dots & \dots & \dots & \frac{df_1}{dx_n} \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ \frac{df_n}{dx_1} & \dots & \dots & \dots & \frac{df_n}{dx_n} \end{bmatrix} \quad (11)$$

Where $f(\dots)$ can be evaluated as in the equation (11).

5. Simulation Results:

For validation of proposed PID with EKF control technique in fixed-wing UAV's roll angle stabilization, MATLAB plate form is chosen. The parameters of PID controller are chosen by auto-tune method in such a way that the roll stabilized as fast as possible. These values are $K_p = 0.4593$, $K_i = 6.4651$, $K_d = -0.0123$ with $N = 37.3396$. The Fix-wing UAV Roll angle stabilization model parameters are chosen as mention in Table-1.

Figure 3 shows the open loop step response of Roll movement of UAV which shows that its amplitude is totally uncontrolled. Figure 4 shows the close loop response of Roll movement with PID controller, in the presence of sensor noises which is added in its feedback loop. This figure validates the effectiveness of PID controller, it quickly maintained the desired roll angle value, but due to the presence of noise in its feedback loop, the output is noisy and not so good. Figure 5 shows the noise signal which is added to system with variance of 0.5m/s and amplitude of 0.1. Figure 6 shows the complete effectiveness of proposed PID with EKF algorithm which gives desired output response with elimination of unwanted noises for Roll angle stabilization of Fix-wing UAV.

K1	0.0046
K2	0.0046
K3	0.0211
K4	1.5586e-4
K5	0.0015
K6	0.0022
K7	0.0076
K8	6.6266e-5

TABLE I
System Coefficients

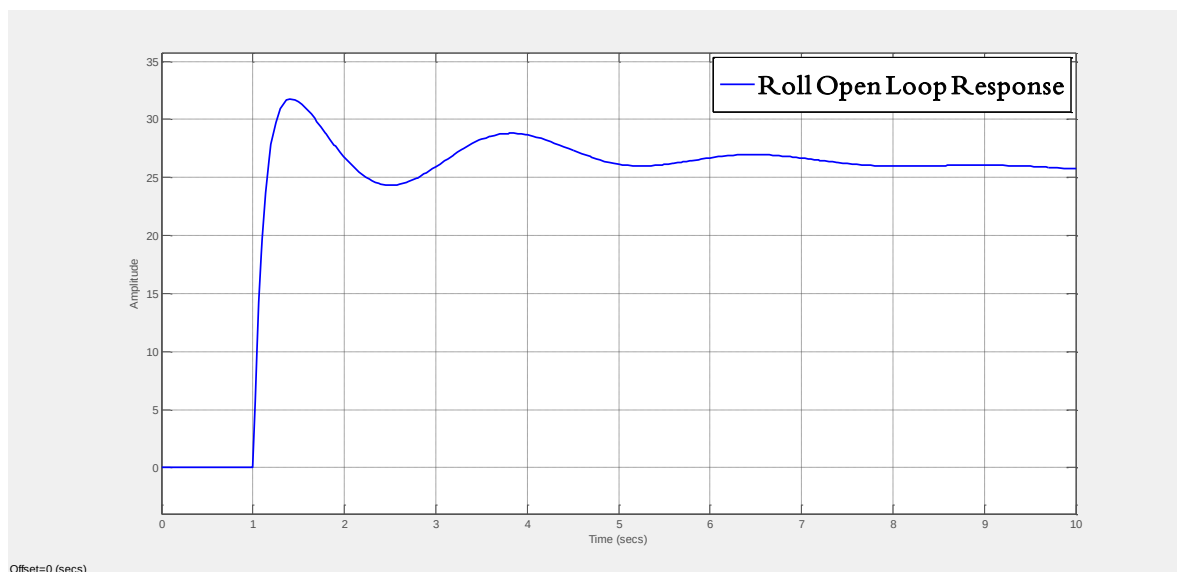


Figure 3: Roll Open Loop Response

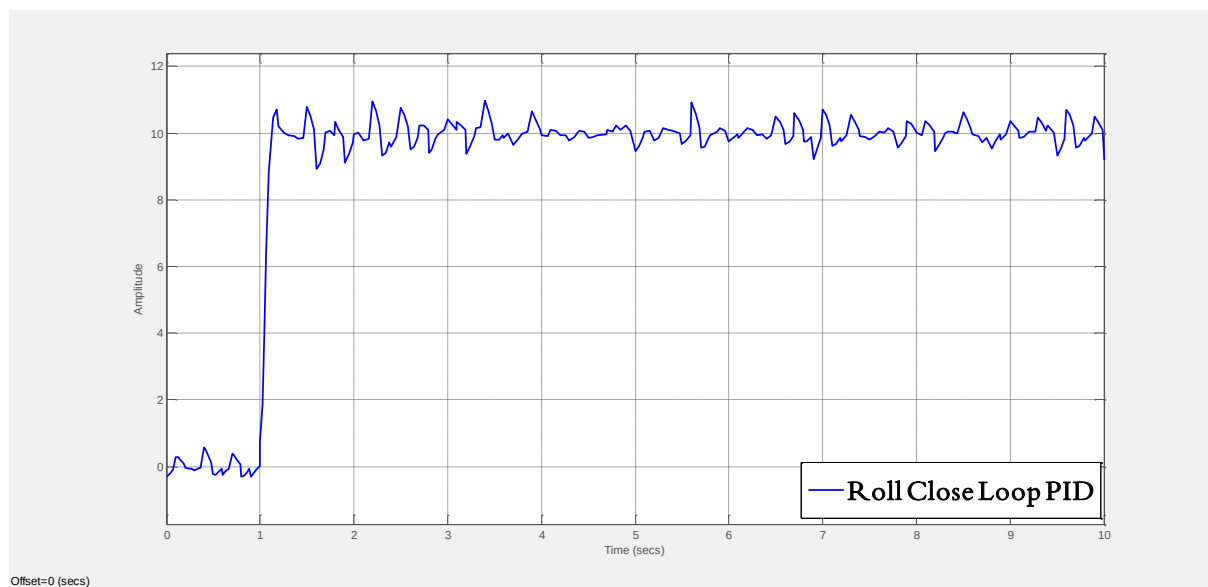


Figure 4: Roll Close Loop PID Response with Sensor Noise

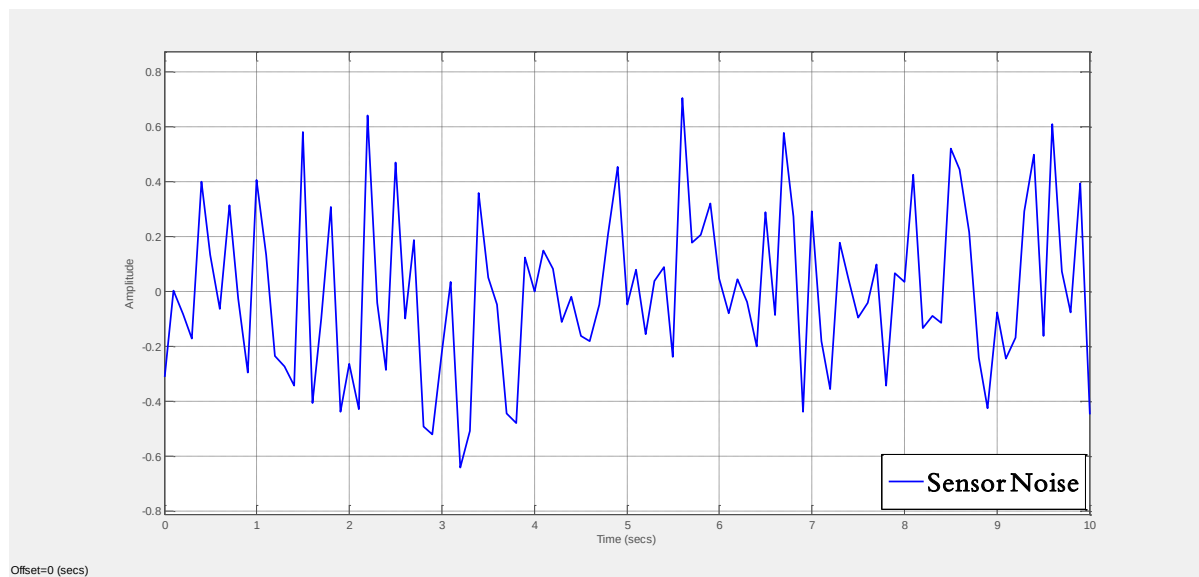


Figure 5: Feedback Sensor Noise

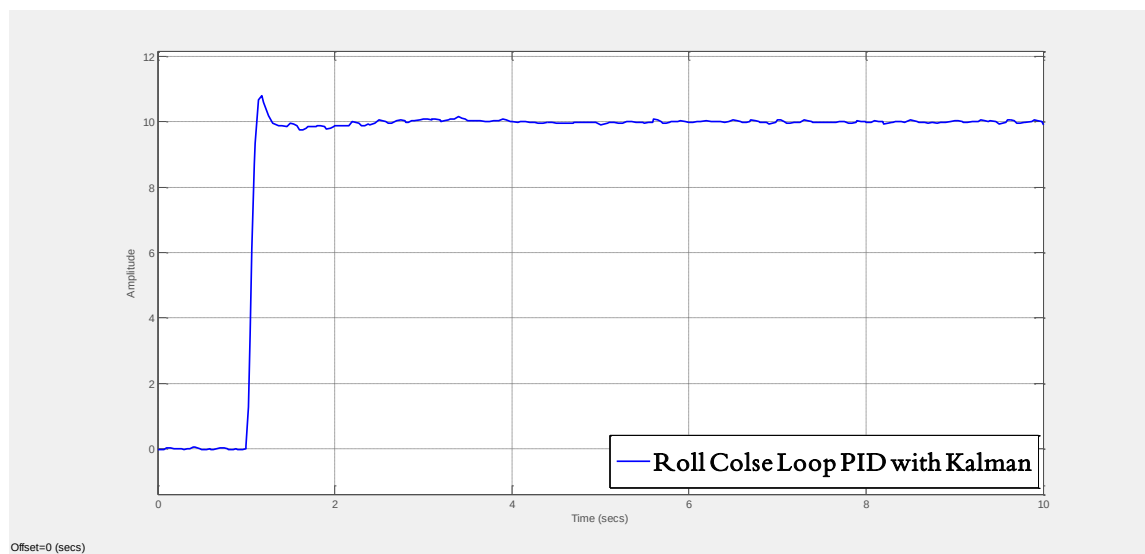


Figure 6: Roll Close loop PID Response with Kalman Filter

6. Conclusion:

This paper presents the design of roll movement stabilization controller for fixed-wing UAV. Design of controller is based on PID with Extended Kalman Filter technique. According to the simulation results it is observed that PID controller stabilize the roll movement but under noisy conditions its output is disturbed, but after combining PID controller with EKF, it gives very good results. From the simulation results we can conclude that the proposed PID with EKF control methodology is good enough for the Roll movement stability under noisy and disturbed conditions.

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